



Short bowel syndrome: epidemiology and etiology

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KEYWORDS

Short bowel syndrome;
Intestinal failure;
Pediatrics;
Epidemiology;
Etiology;
Incidence

Pediatric short bowel syndrome (SBS) is most commonly caused by congenital or acquired conditions of the newborn. SBS is associated with an inability of the bowel to adequately absorb water and nutrients in sufficient quantities to meet caloric, fluid, and electrolyte demands, thus necessitating dependence on parenteral nutrition (PN). It is this dependence on PN, that is responsible for the majority of morbidity and mortality associated with SBS, including central venous catheter infections and PN-induced cholestatic liver dysfunction. There are very few estimates of SBS incidence and mortality in the literature. The epidemiology of SBS is reviewed and the limitations of the published literature are discussed.

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Short bowel syndrome (SBS) is defined as the spectrum of malabsorption that occurs after resection of a major portion of the small intestine for congenital or acquired lesions.^{1,2} It is also defined as the need for prolonged parenteral nutrition (PN) secondary to intestinal failure after bowel resection.³ These patients suffer a multitude of complications secondary to long-term hospitalization and prolonged PN. The earliest descriptions of SBS in the published pediatric studies cite midgut volvulus and intestinal atresia as the most common etiologies, but more recent data suggest that NEC is the largest contributing agent.⁴⁻⁶ In older children and adults, multiple resections for Crohn's disease can be implicated, as can long-segment resections for ischemic vascular disease, radiation enteritis, malignancy, trauma, and adhesive obstruction.⁷ Other causes for intestinal failure include intestinal motility disorders and mucosal enteropathies.⁸

Regardless of the etiology, massive intestinal loss and compromised bowel adaptation lead to a predictable sequence of secondary events. The multitude of complications that ensue with long-term hospitalization and prolonged PN^{9,10} include central line complications, multiple systemic infections,

cholestasis, liver failure, and failure to thrive.^{5,6,11,12} Prompt weaning from PN seems necessary to prevent liver dysfunction and improve outcome. As PN-induced liver failure progresses to cirrhosis, small bowel and liver transplant become the only therapeutic option for these infants. Many of these infants do not survive long enough to receive a bowel and or liver transplant as they are too premature and small to be eligible for the procedure.¹³

Accurate estimation of SBS incidence and mortality remains difficult due to variation in the SBS definition between studies, lack of comprehensive follow-up, and inability of tertiary institutions to clearly define their study population because of complex referral patterns. Estimates of incidence and mortality would be helpful in enabling health care personnel to appropriately counsel parents, allocate resources, and plan clinical trials. Identifying the factors associated with prolonged dependence on intravenous nutrition, development of liver disease, or death is crucial, particularly if these factors are amenable to changes in clinical care.

Classification and etiology

Intestinal failure is a comprehensive term describing a malabsorptive state in which the residual intestinal function is inad-

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equate to support growth and hydration. Intestinal failure may be caused by SBS, mucosal enteropathy, or dysmotility syndromes. Some patients possess elements of 2 or all 3 categories (ie, gastroschisis). The most common cause of intestinal failure in pediatrics is SBS. In most cases, gut loss occurs because of congenital or acquired conditions of the newborn.

SBS can be classified into 3 anatomical subtypes: (1) Small intestine resection with a small bowel anastomosis and intact colon; usually, some ileum is preserved. (2) Small bowel resection with partial colon resection and resulting enterocolonic anastomosis. (3) Small bowel resection with high-output jejunostomy. Type 1 is the best tolerated subtype with the most optimum adaptive potential. Interestingly, it is also the subtype that is most represented in animal models of SBS. Type 2 is more commonly encountered clinically as NEC commonly affects the ileum and right colon. Type 3 is the most challenging subtype to manage due to the high fluid replacement requirements. Massive jejunal resections, on the whole, are better tolerated than large ileal resections.^{7,14} Massive ileal resections tend to result in an inability to resorb both dietary fluid and fluid secreted by the jejunum to maintain iso-osmolality with the enteric lumen. The tight junctions in the jejunal epithelium are more “leaky,” allowing free transit of fluid and salt across the mucosa to maintain an intraluminal concentration of 70-90 mmol/L. The ileal mucosa is much “tighter” with a net absorption of fluid and salt. Additionally, these resections are associated with impairment of resorption of vitamin B₁₂, bile salts, and fatty acids. In turn, disruption of the enterohepatic circulation and the delivery of unabsorbed bile salts into the colon stimulate colonic secretion and promote motility. In time, the colon may adapt to the delivery of large volume of solutes.¹⁵⁻¹⁷ It will also augment energy absorption through the metabolism of unabsorbed carbohydrate into short-chain fatty acids by resident bacteria.^{18,19}

There is a clear correlation between intestinal length and patient outcome, but there is much more to the equation than absolute bowel length. The region of intestine remaining, the primary diagnosis, the functional capacity of the residual intestine, the presence of colon, and the age of the patient are all relevant. Nearly 4 decades ago, Wilmore suggested that survival with SBS requires 15 cm of small intestine in the presence of the ileocecal valve (ICV) or 40 cm without it.²⁰⁻²² These numbers have been challenged by recent data; however, predictably, mean small bowel length (SBL) and colon length always emerge to the forefront of any discussion relating to the development of SBS, both in absolute terms and when expressed as percentage of original bowel length. We have previously highlighted the importance of expressing residual intestinal length as a percentage of the predicted length.⁵ Spencer et al reported that a SBL of <10% of the expected intestinal length is associated with a relative risk of 5.74 for death.¹¹ Although absolute bowel length was predictive of this adverse outcome, the magnitude of risk was lower. One of the limitations of using absolute remaining intestinal length relates to developmental changes, par-

ticularly in a population where prematurity is common. Bowel is known to undergo a rapid increase in size between 30 and 40 weeks of gestation, lengthening by nearly 150%.^{23,24} SBL nearly doubles again in the first 18 months of life.^{23,25} The development of SBS was also associated with a statistically higher proportion of jejunostomies, ileostomies, and ICV resections, suggesting that development of SBS may be associated with resection of distal small bowel (Wales⁵). Data are conflicting regarding the importance of preservation of the ICV in prevention of SBS.^{5,11,12,26,27} Most contemporary studies have found that, when controlling for length of resection, the presence or absence of an ICV does not have a significant effect on the subsequent development of SBS. Rather, the absence of an ICV may be a marker simply for a larger resection, as well as, a marker for resection of ileum, which is considered to have greater ability to adapt and to increase net absorption than more proximal intestinal segments.

Quantification of normal intestinal length in neonates and infants has received little attention in published studies. Studies have measured the intestinal length using a variety of approaches, thus generating results that are highly variable. Measurements of small intestinal length in a full-term neonate vary between 176 and 305 cm. However, all these values represent postmortem measurements.^{11,25,28-34} In adults, the small intestinal length ranges from 2.4 to 3.7 m when measured in vivo, with a range of 4.7-9.7 m postmortem.³⁵ These discrepancies are likely due to alterations occurring in the perimortem period, with early shortening of the intestine as a result from smooth muscle contraction. However, as autolysis proceeds, relaxation and lengthening of the intestine occurs.³⁶ The measurement technique is also important when interpreting intestinal length, as tension may increase length in the small intestine by 4% and in the large intestine by 14%.³⁷ Due to the limitations of existing postmortem data, which have been used to date for the quantification of intestinal length, our group sought to examine intestinal length in live infants.²³ Measurements were done in infants from birth to age of 5 years. We also examined weight and height at surgery as separate predictors of intestinal length. Our data suggested that height at surgery (in centimeters) be used to estimate SBL using the following equation: $\ln(\text{SBL}) = 6.741 - 80.409/\text{height}$. Similarly, colon length may be predicted using the following equation: $\text{CL} = 0.111 \times \text{height}$.^{1,521} We chose height for several reasons. First, height was most predictive when we examined the variability from observed to expected values across the entire spectrum of height. This is in contrast to age and weight that had minimal variability at the bottom range of the spectrum and significantly more variability in bowel length as an infant aged or gained weight. Second, height was easy to quantify and was relatively stable with time, and was not subject to drastic changes in the immediate perioperative period as weight was. Finally, gestational age may be more difficult to accurately quantify than height (Figures 1 and 2).

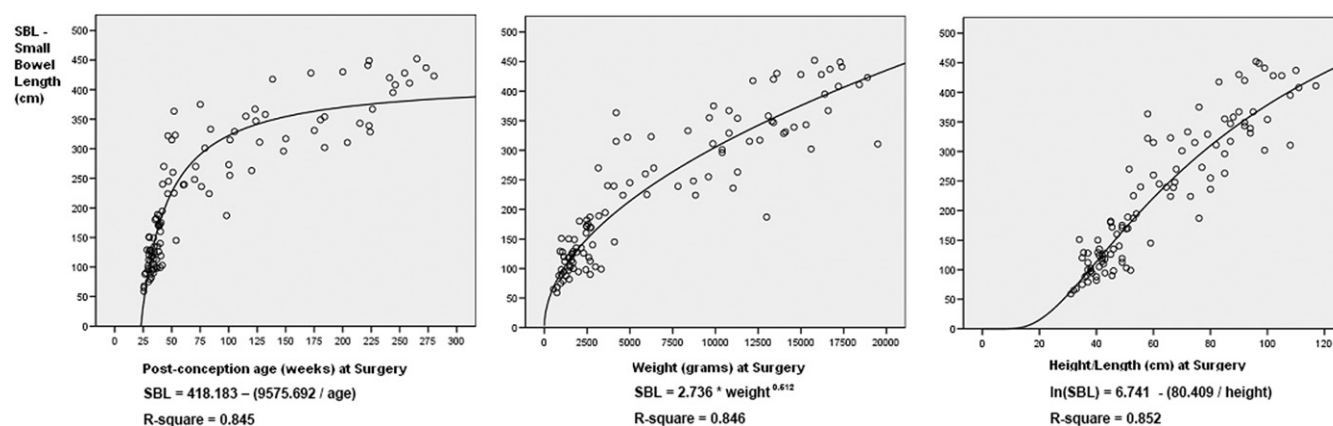


Figure 1 Graphs, based on optimal model from curve fit process, for prediction of small bowel length from each of postconception age, weight, and height at operation. (Reprinted with permission.²³)

The main causes of SBS in children remain congenital and perinatal diseases, such as intestinal atresia, abdominal wall defects (gastroschisis), malrotation/volvulus, and long-segment Hirschsprung's disease.^{6,11,38} The most common cause, however, is necrotizing enterocolitis (NEC) making up at least 30% of most reported series. The number of patients with SBS secondary to NEC is expected to increase as more extremely premature infants are resuscitated and survive the initial weeks after birth. In addition, as more infants with complex congenital heart disease are salvaged, an increasing proportion of patients with ischemic intestinal complications will be encountered.

Approximately, 20% of SBS in the pediatric population develops outside of the neonatal period.¹¹ Volvulus and trauma are the leading etiologies in the non-neonatal population. Crohn's disease was conventionally a significant cause due to bowel loss from recurrent resections. Improved understanding of the pathogenesis of Crohn's disease and the development of novel therapies, such as immunomodulators has resulted in a reduction in the frequency and extent of surgical resections. Crohn's disease, therefore, may be-

come a less prevalent cause of SBS mesenteric infarction from thrombotic disease and malignancy are other etiologies seen more in the adult population than in pediatrics.

Incidence and mortality estimates

Accurate estimates of neonatal SBS incidence and mortality are difficult to determine due to the rarity of the condition, variation in the definition of SBS between institutions, difficulty of tertiary care referral centers to accurately determine their catchment population and problems ensuring complete follow-up of the cohort. Also, inability to clearly define the population at risk reduces the reliability of the estimates. Newborns and infants, not adults, are at risk of neonatal SBS. Incidence risks based on the whole population, for example, the Canadian Collaborative Study Group estimate of 4.8/1 million population/year, can be misleading.³⁹

Usually, the only mortality estimates published have been the equivalent of the case fatality rate. The case fatality

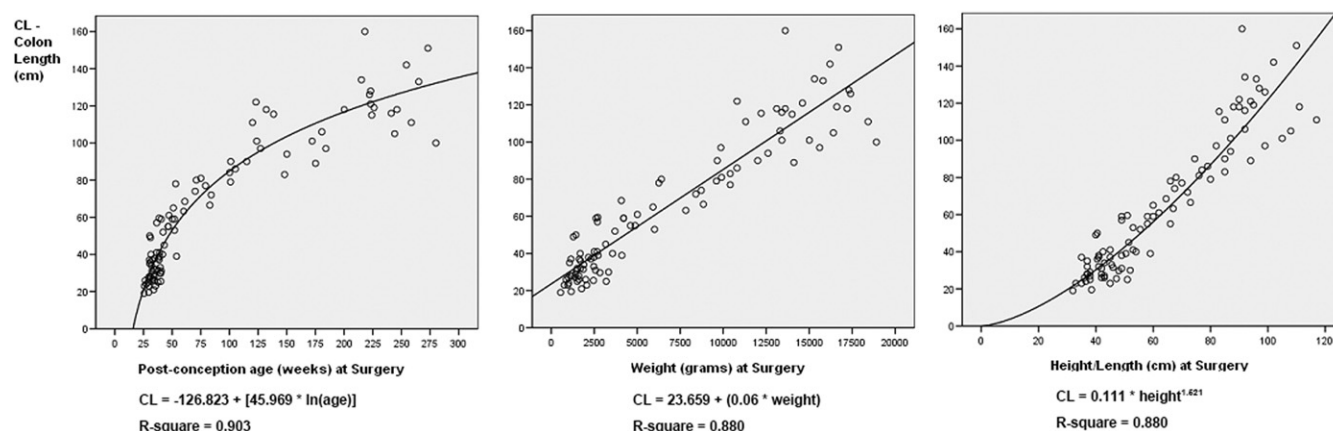


Figure 2 Graphs, based on optimal model from curve fit process, for prediction of colon length from each of postconception age, weight, and height at operation. (Reprinted with permission.²³)

rate is the number of deaths from a disease divided by all cases of that disease. It represents the proportion of fatal cases among affected individuals. The case fatality rate is a measure of disease severity.⁴⁰ For mortality estimates to be accurate in observational studies an adequate follow-up period must have occurred to allow enough time for the event to happen. Also, all participants need to be followed up to prevent observation bias.⁴¹ In observational studies, the length of follow-up is different for all study participants as they are recruited at different times. Cumulative survival estimates ensure that all available follow-up data are utilized in the calculations.

Epidemiologic data about the prevalence or incidence of intestinal failure in pediatrics is sparse. The prevalence of home PN has been reported as a proxy for intestinal failure incidence, but these data do not include patients with intestinal failure with a temporary need for parenteral support. Also, some patients on home PN may require it for nutritional supplementation for treatment of malignancy, etc., and do not accurately represent patients with true intestinal failure. European surveys of home PN indicated an incidence of 2-3 patients per million and a prevalence of 4 per million.^{42,43} As home PN management has become progressively more decentralized, few studies on incidence of home PN in Europe have been published recently. This makes estimation of epidemiologic data for intestinal failure even more difficult because some proportion of patients are unaccounted for, or may be weaned off PN before entering home PN programs. Messing et al reported that as many as 27% of patients with SBS weaned off PN 1 month after the event that caused intestinal failure. In the United States, larger figures have been reported, which imply a different indication or philosophy for this treatment between Europe and North America.^{44,45}

Cole et al published data examining incidence and outcome of SBS in very low birth-weight infants from a consortium of 16 tertiary neonatal centers in the United States. The incidence of surgical SBS was 0.7% (7/1000) in 12,316 very low birth-weight infants and 1.1% (11/1000) among 5657 extremely low birth-weight infants. NEC was responsible for 96% of cases. The data represent the largest cohort of neonates assessed for the incidence of SBS across the United States. The estimates do not include term infants where congenital anomalies play a much bigger role in etiology. Unfortunately, SBS was not defined beyond a need for PN support for malabsorption after gastrointestinal surgery with significant bowel resection.⁴⁶

In 2008, the *Italian Society of Paediatric Gastroenterology Hepatology and Nutrition* and the *Italian Society of Neonatology* published data pooled from 7 tertiary neonatal intensive care units over 2003 and 2004.⁴⁷ SBS was defined using the definition employed by our program (a residual SBL <25% of that predicted for gestational age or the need for PN for more than 42 days after intestinal loss/surgery). To determine incidence estimates, the number of live births

and admissions during this study period was provided by each center. The rate of intestinal failure was 0.1% in all live births (26/30,353) and 0.5% (26/5088) among neonatal intensive care unit admissions.

The Canadian Collaborative Study Group³⁹ estimated the incidence of SBS across Canada as 4.8/million population/year, but the calculation was based on a sample size of only 11 infants. This was partly secondary to poor contribution from larger institutions with the critical mass of patients. Studies examining the need for small bowel transplant have suggested an incidence of 2-3 patients/million population/year, of which half are pediatric.¹³ Mortality rates vary tremendously or are not presented at all. Because patient selection often excludes early deaths, there could be an overestimation of SBS survival in published reports.

To our knowledge, only a single population-based study has been published, which examined SBS incidence and mortality. This Canadian study used administrative health care data and federal census data within a defined catchment area to determine estimates.⁴⁸ The incidence of neonatal SBS was found to be 24.5 per 100,000 live births (95% confidence interval (CI) = 12.1, 36.9) with a much higher incidence in infants born before 37-week gestation compared with term newborns (353.7/100,000 live births vs 3.5/100,000 live births). The most common etiologies among this age group were NEC (35%), complicated meconium ileus (20%), abdominal wall defects (12.5%), intestinal atresia (10%), and volvulus (10%). We found the overall mortality rate of neonates with SBS to be 37.5%, which is 3 times that of a cohort of infants matched by underlying diagnosis but without SBS.^{5,48} Sixty percent of SBS-related deaths were attributed to hepatic failure. Other series report similar mortality rates of 20%-40%, which have remained relatively constant during the past several decades, despite advances in neonatal care.^{4,49,50} In our historical experience, mortality in SBS was bimodally distributed.^{5,12} These results were explained by the definition of SBS that we chose for this study. The patients who died shortly after surgery were typically the ones who had small bowel resection of greater than 75%. The anatomical component to our definition captured this subgroup of patients and may have increased our mortality estimates relative to other studies. In patients who survived the initial procedure, a second period of increased mortality of 8-12 months before survival plateaued at 63%. The late deaths were attributed to sepsis and intestinal failure-associated liver disease. In fact, a recent study from University of Michigan found that there were no late deaths in patients with SBS who were able to be weaned off PN.¹¹ A similar study from Pittsburgh cited 5-year survival of 95% in SBS patients who were able to be weaned off PN vs 52% in those not weaned.⁵¹ While large-scale data collection is lacking, recent developments in the prevention and treatment of intestinal failure-associated liver disease will markedly decrease the mid- to long-term mortality associated with SBS.^{6,52,53}

Limitations of published SBS data

Accurate predictions of SBS incidence have not been possible as most of our knowledge is based on level III evidence from single-institution case series. In 2004, we outlined shortcomings of the published data on SBS,⁴⁸ and although there have been some improvements, the issues that remain pertinent today are as follows:

Participant selection and study design

Due to the small number of patients treated at individual institutions, it is often necessary to base our management on level III evidence that may be flawed in study design, patient selection, definition, and statistical analysis. It is difficult to interpret results and critically compare studies to one another because of conflicting definitions of SBS, varying demographics, a myriad of management practices, inconsistent stratification for gestational age, and lack of control groups.

Most articles fail to clearly state how subjects were selected. Case series occasionally mention that “consecutive cases” were enrolled. Inclusion and exclusion criteria are often not mentioned and there is usually no clearly defined control group. Selection bias is possible. Bias represents a systematic error that results in incorrect estimation of the association between exposure and disease.⁴⁰ Every analytical study is susceptible to bias and it is up to the reader to decide whether bias distorts the results sufficiently to disregard the conclusions.

Selection bias is an error in the process of identifying the study population.⁴¹ It is a particular problem in retrospective cohort studies because the exposure and outcome have already occurred at the time of participant enrolment. Surgical research is particularly vulnerable to selection bias because surgical indications may be subjective and governed by a surgeon’s judgment. Even when guidelines for operation are clear (as in interventional trials), an individual surgeon may or may not decide that surgery is indicated. Thus, a study could be criticized because the surgical cohort may not represent the true population. One advantage of pediatric SBS, in which the etiology is usually based on congenital diseases of the newborn, is that the indications for surgical intervention are usually clear. The exception, however, is the population of newborns who have NEC. Indications for surgery in this group are more subjective. The clinical decision to operate could directly affect the number of study participants. Typically, neonates with known abdominal pathology do not receive surgical therapy only if their comorbid disease precludes surgical intervention. This potential selection bias needs to be considered in all surgical research and interpreted to determine whether the results are still valid.

Observation bias relates to differences in the way exposure or outcome data are obtained from study groups.⁴⁰ Abstraction of data retrospectively and incomplete follow-up will affect results.

Long recruitment time

Many of the published series have a recruitment time spanning more than a decade.⁴⁸ This is in part related to the small number of patients treated at any 1 institution. The problem, however, is that management of the critical care neonate would be expected to advance over a 10- to 15-year period with improvements in PN, mechanical ventilation, and surgical technique. Therefore, including “historical” patients in an outcome analysis does not reflect current medical practice and introduces bias.

Definition of SBS

There continues to be no consensus in the literature for the definition of SBS. In general, there is agreement that SBS exists when a patient’s residual intestinal function is inadequate to sustain survival, growth, hydration, and electrolyte homeostasis. Beyond this general principle, no strict criteria exist. Variability of the SBS definition among studies makes comparison of results between institutions difficult and, in many cases, SBS is not defined at all.

SBS may be defined anatomically based on the amount of residual small intestine remaining or by duration of PN dependency. It is our belief that SBS is essentially a functional disorder. An anatomic definition based only on residual length of small intestine is not reliable due to numerous confounders, such as patient age, region of bowel remaining, underlying diagnosis, and presence or absence of the colon that will affect intestinal adaptive potential. Alternatively, a definition of SBS based solely on length of postoperative PN (eg, >3 months) selects for survivors because patients who die early in the postoperative course after massive small bowel resection, are excluded. Therefore, an important subgroup is excluded from analysis and morbidity and mortality estimates may be underestimated. The definition adopted by our multidisciplinary intestinal rehabilitation team in 2002 was originally proposed by the Canadian Association of Pediatric surgeons.⁵⁴ It remains primarily a functional definition with patients who remain PN dependent for greater than 6 weeks, but it also incorporates an anatomical component, including patients with massive small bowel resection of greater than 75%.

Failure to account for gestational age in bowel length estimates

Consideration of gestational age is imperative when reporting residual bowel length. Bowel length increases throughout gestation with a rapid increase between 30 weeks and 6 months postnatally. Unlike adults, infants and children possess much more gut growth potential due to ongoing developmental processes. Therefore, to understand the relevance of residual intestinal length, one must know what normal gut length is. In the commonly cited article by Touloukian in 1983, intestines were measured in 30 stillborn infants

divided into 3 broad gestational age groups ranging from 24 to 40 weeks. He found that a 28-week-old fetus has approximately 200 cm of bowel and this increases to 300 cm at term.²⁴ Our group recently published norms for both small and large intestinal length for premature infants of 24 weeks to children of 5 years²³ on the basis of data obtained from a large cohort of live patients. We found that a term infant has approximately 200 cm of small intestine and 40-50 cm of colon. Our results provide normal reference values. Hence, the outcome of a 28-week gestation infant with 30 cm of residual intestine may be different from that of a term infant with 30 cm of residual intestine because of the gut's ability to grow in length as the infant matures. Therefore, studies that report residual bowel length as an absolute measurement (in centimeters), without adjusting for gestational age, are misleading. To account for gestational age, residual bowel should be presented as a percentage of the original total bowel length or of that expected for a particular age.

In summary, although SBS is rare in the population at large, it comprises a sizeable proportion of NICU admissions and still carries a significant risk of mortality, especially among premature infants. As neonatal intensive care moves toward sustaining an increasingly premature population of infants, a population that is at greatest risk for development of NEC, we may predict that the incidence of SBS will increase. The institution of multidisciplinary care teams, novel surgical procedures (ie, serial transverse enteroplasty) and changes to the lipid component of PN hold great promise for reducing PN-related mortality and improving overall outcome.

Presently, despite superior results, neonatal patients with SBS suffer tremendous morbidity and mortality. They require a significant intellectual, emotional, and financial investment. Accurate estimates of incidence and mortality of this condition are sparse. Unfortunately, current published data are limited primarily due to research design and lack of a standardized definition. Population-based epidemiologic data will assist clinicians in counseling parents, allocating resources, and planning clinical trials. To advance the field, the focus needs to be on improving research methodology. Due to the relative rarity of the condition at individual institutions, large-scale multicenter initiatives, for example, research consortiums or data registries such as those that exist for intestinal transplantation, would be a step in the right direction.

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